

However, quantum computing is not without its challenges. One of the most pressing issues is the high error rate in quantum systems. Quantum computers are incredibly sensitive to their surroundings; even small changes in temperature or electromagnetic interference can cause errors. Researchers are working on quantum error correction techniques, but creating stable, large-scale systems remains a complex task. Despite this, the promise of quantum computing remains strong, and we are making strides toward overcoming these obstacles.

Experts predict that it may take a decade or more before quantum computers are fully commercialised and accessible to the public. Until then, we might see hybrid systems, where quantum computers work alongside traditional computers to solve specific, computationally intensive problems.

As quantum computing progresses, it will also raise important ethical and societal questions. If quantum computers become powerful enough to break current encryption systems, our data could be at risk. To address this, quantum-safe encryption is already being researched, and countries, including India, are developing policies to ensure that we stay ahead of potential threats. Additionally, as the demand for quantum technology grows, we will see new job opportunities emerge in quantum programming, engineering, and ethical data handling.

In India, leading institutions such as the Indian Institutes of Technology (IITs) and Indian Institute of Science

(IISc) are offering specialised programs to prepare students for this future. Similarly, top global universities, including MIT and Oxford, are expanding their quantum computing courses to meet the growing demand. This new wave of education presents a unique opportunity for students to become pioneers in this field.

Quantum computing has the potential to solve some of the world's toughest problems. Although it is still in its early stages, its impact is already beginning to show, and the possibilities are endless. For youngsters, it's a chance to be part of a technological revolution that will shape the future. **END** 

 **By: Dr Chinmoy Kumar**

The author is an associate professor in business analytics at SCMS - PG, Dayananda Sagar University, Bengaluru. His research and teaching interests include analytics, data science, emerging technologies, business models, and e-governance for sustainable development. He is also the founder of The Learning Network, a pro bono educational venture based in Ranchi, Jharkhand, India.

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- **User-driven features:** The development team actively reviews community suggestions and feedback, prioritising features that users identify as beneficial for their research and applications.

Within the broader quantum computing ecosystem, Pytket occupies a vital position as a toolkit that bridges the gap between theoretical research and practical application. By advancing accessible and efficient quantum computing, Pytket enables a diverse range of users—from academic researchers exploring new quantum algorithms to industry professionals implementing solutions in finance, logistics, and beyond. Its commitment to open source development fosters

community collaboration, driving innovation and enhancing the toolkit's capabilities over time.

As quantum computing continues to evolve, Pytket represents an invaluable resource for those looking to experiment with and apply quantum technologies. Whether you are a seasoned quantum researcher or a newcomer exploring this

exciting field, Pytket provides the tools necessary to experiment, prototype, and implement quantum algorithms effectively. With its growing role in the quantum development landscape, now is an excellent time to explore Pytket and discover the potential it holds for shaping the future of quantum computing. **END** 

Reference

- <https://github.com/CambridgeQuantum/pytket>

 **By: Sufyan bin Uzayr**

The author, apart from being a writer, is an educator and entrepreneur. He holds multiple degrees in management, information technology and humanities. He has written seven books so far and is currently the director of a multinational IT company. As an educator, he has taught at universities and institutions worldwide.